European Polymer Journal 222 (2025) 113589

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

NIR photothermal activation in epoxy/anhydride systems for advanced polymerization

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A B S T R A C T

Photopolymerization is an important area of investigation due to its wide industrial applications; however, it generally employs quite energetic UV or near UV light to initiate the formation of the polymer. Even though it is very effective and offers several advantages, numerous drawbacks are also encountered such as low light penetration, harmfulness of wavelengths in the UV range and the use of photoinitiators that can be hazardous to health. In this context, many researchers have focused their interest in improving the spatial localization and the temporal control of the polymerization under safer irradiation wavelengths (NIR). In this work, we have successfully produced polyester thermosets by the reaction of epoxy resins with anhydride in the presence of a thermolatent *N*-Heterocyclic carbene generator using the NIR photothermal effect of a dye (heater) able to convert low-energetic NIR light into heat.

1. Introduction

Photopolymerization is a room temperature process allowing both temporal and spatial control of the polymerization [1] that finds application in various areas such as coatings, inks, adhesives and additive manufacturing (3D printing) [2]. Nevertheless, most of the photopolymerization processes are produced upon UV and near-UV irradiation that present certain drawbacks. Firstly, the limited photon penetration in UV wavelengths restricts the production of thick or composite materials [3]. Secondly, UV light is harmful to the skin and the eyes, which poses a safety problem for operators. Additionally, formulations containing UV/Vis-sensitive compounds must be handled in the dark, limiting their use in certain applications and resulting in coloration of the final material.[4].

As a consequence, there is a growing demand for a technology offering spatial and temporal control of polymerization but that uses safer irradiation wavelengths such as near-infrared (NIR). Light of such wavelengths (larger than 780 nm) also has a deeper penetration in materials.[5] We will present the activation by NIR light of the AB polyaddition reaction, but the development of light sources (and their implementation in industrial environments) is also motivated by the possibility of using this light for other applications (not only based on polymerization, coatings or additive manufacture). Such are for example photopolymerisation of acrylate system by using a radical generator,[6] by up-converting of the energy of the light, [5] metal-free RAFT in aqueous conditions (acrylates),[7] thermal evaporation (physical drying) and film formation for coatings… Moreover, if this technology would be able to induce an increase of the temperature of the reaction medium, it would provide spatial and temporal control to most conventional thermal polymerization reactions developed to date.[8].

In this context, the photothermal effect has emerged in recent years as a promising strategy for converting NIR radiation into heat, offering high spatial and temporal resolution as well as significant penetration depth [9]. This effect enables the generation of heat from NIR radiation with high precision and can provide an excellent light penetration over several centimeters, allowing the potential formation of polymer materials in thicker layers.[10] Thus, the idea of combining the advantages of photopolymerization and the photothermal effect is particularly appealing for developing new polymerization strategies. The use of commercial dyes as heaters in photothermal free radical polymerizations but also for thermoplastic polymer reprocessing has been demonstrated in recent preliminary experiments.[11] When irradiated at 1064 nm, temperatures up to 250 °C could be achieved. However, a match between the NIR light emission wavelength and the absorption maxima of NIR heaters is necessary to achieve these temperatures.[12]

While epoxides as single monomers can be used in three

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https://doi.org/10.1016/j.eurpolymj.2024.113589
Received 28 September 2024; Received in revised form 13 November 2024; Accepted 18 November 2024
Available online 23 November 2024
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predominant classes for their polymerization mechanism which are metal-mediated, cationic and anionic homopolymerization,[13] epoxides are widely used in polyaddition reactions with "so called" hardeners resulting in AB(AE) polymers. For example, epoxy(E)-amine(A) systems predominate due to their abilities to harden at moderate temperatures [10]. With anhydrides as second monomer (hardener), higher polymerization temperatures are required, but on the other hand materials with higher glass transition temperatures (T_g) to those from epoxy/amine systems can be reached. Hence this kind of polymer has a large number of challenging and important applications also in the field of polymeric coatings, electronic devices, adhesives, binders for prepregs [14] and composite material like for impregnation of stator winding of large rotating electrical machines.[15] This latter epoxy/anhydride process demands greater thermal activation and is generally less efficient.[16] Moreover, in the authors' opinion, epoxy/anhydride systems are particularly interesting due to the backbone of the polymer material containing ester functions that can be degraded under basic conditions, thus increasing the potential recyclability of such materials [17]. Existing OH groups in the resin and a number of different "accelerators"[15] are in a complex manner beneficial for the processing / reaction leading to the polymer.

The possible activation of epoxy/anhydride systems by light to speed-up the reaction and reduce the energetic cost is a huge challenge for these widely used polymers e.g. by reducing or preventing additional heating for curing and/or postcuring. To achieve this objective in our study, NIR photothermal ring-opening copolymerizations (ROCOP) of epoxy/anhydride systems were carried out in the presence of NIR dyes as heaters and a *N*-heterocyclic carbene generator (NHC-G) used as thermolatent initiator. Markedly, the heat can be directly generated in the material from the NIR dye irradiation through a photothermal effect allowing a better spatial and temporal control of the heat exposure (on/off) that is not possible to do with an oven; accordingly, a high efficiency process is found (curing in 300 s). The polymerization processes were followed by monitoring temperature with a thermal imaging camera, and by tracking the conversion over time by Fourier-transform infrared spectroscopy (FTIR). All the NIR heaters were investigated by thermogravimetric experiments (TGA) to characterize their thermal stabilities. The final polymer properties (T_g and residual epoxy content) were investigated by differential scanning calorimetry (DSC) experiments.

2. Experimental section

2.1. Chemical compounds

2,2′-[(1-Methylethylidene)*bis*(4,1-phenyleneoxymethylene)]*bis*oxirane (BADGE) and tetrahydromethyl phthalic anhydride (THPA) were obtained from Merck/Sigma-Aldrich with the highest available purity and used as epoxy monomers and hardeners, respectively (Fig. 1). Note that we can assume a far lower amount of –OH groups as for commercial BADGE based resins (higher purity). The *N*-heterocyclic carbene generator 1,3-Dimesitylimidazolium-2-carboxylate (NHC-G) as shown in Fig. 2 was used as thermal initiator (TCI-Europe; purity > 95.0 %). The different used near-infrared (NIR) dyes (Fig. 3 and Table S1) were obtained from Sigma-Aldrich (Europe), Lotchem Ltd (China) and Few Chemicals (Germany). The maximal absorption wavelength and decomposition temperatures of the dyes were measured and presented below in the results and discussion section.

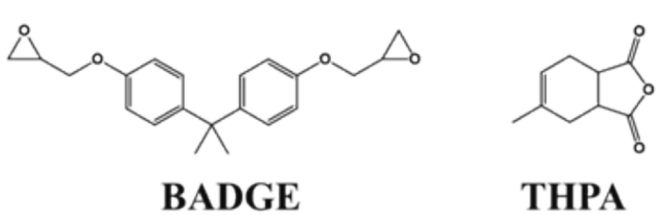


Fig. 1. Epoxy and anhydride monomer used in this work.

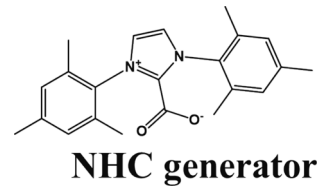


Fig. 2. Chemical structure of the thermolatent *N*-heterocyclic carbene generator (NHC-G), 1,3-Dimesitylimidazolium-2-carboxylate employed in this study.

2.2. Epoxy/anhydride polymerization procedure

The stoichiometric ratio between epoxy and anhydride functionalities was selected to ensure an epoxy/anhydride polyaddition process, assuming that one bifunctional epoxy monomer reacts two times with one anhydride molecule. The choice of this stoichiometric ratio has already been described in detail in our previous work.[18] The polymerizable resin was composed of BADGE/THPA (48 %/52 % w/w).

The percentages of the various chemicals in the initiating systems were determined based on the monomer content (by weight). All formulations were prepared using the following procedure: 1,3-Dimesitylimidazolium-2-carboxylate (0.5 phr) and the dye (0.1 phr) were dissolved in tetrahydromethyl phthalic anhydride (THPA) until the formulation reached homogeneity. Subsequently, 2,2′-[(1-methylethylidene)*bis*(4,1-phenyleneoxymethylene)]*bis*oxirane (BADGE) was added (BADGE/THPA (52/48 % w/w)), and the mixture was stirred for 24 h at room temperature (21–25 °C). Analysis of the performance of the system was monitored for 12 to 18 h after formulation of the resin mixtures (stirred with a magnetic bar).

2.3. Irradiation devices

For the irradiation, two NIR laser diodes were employed, both procured from Changchun New Industries (China). The first is a Laser diode@785 nm with an adjustable irradiance ranging from 0 W to 0.4 W cm^{-2}, while the second is a laser diode@1064 nm with an adjustable irradiance ranging from 0 to 0.5 W cm^{-2}.

2.4. RT-FTIR spectroscopy

A JASCO 4600 Real-Time Fourier Transform Infrared Spectrometer (RT-FTIR) was employed to monitor the conversion of reactive epoxy functions over time upon NIR irradiation. Specifically, the conversion of epoxy was tracked based on the peaks in the near-infrared region (4470 to 4568 cm^{-1}) [19] using the following equation:

$$\boldsymbol{RT-FTIR\ Conversion}\ (\%) = [1-\frac{\boldsymbol{E_t}}{\boldsymbol{E_0}}]\boldsymbol{x}100 \quad (1)$$

with E_0 the initial area of the Epoxy peak and E_t the area of the Epoxy peak at the time t of the irradiation. The experiment is carried out in-situ i.e. the sample position in the FTIR spectrometer cavity (using a horizontal accessory) is not changed and we also monitor one non-changing vibrational band to ensure that the "liquid" does not flow out of the monitored area (a reduction in thickness can be controlled in this way). Based on reproducibility experiments the error bars was typically 2 % on the conversions given.

Polymerization was conducted in a mold with a thickness of 1.4 mm, under air, and at room temperature, upon irradiation with a NIR laser diode. The light was activated 10 s after the initial spectrum measurement. Further details regarding the experimental conditions for photopolymerization experiments are provided in the figure captions. The method employed for tracking the photopolymerization profiles has been previously outlined in detail in [20,21].

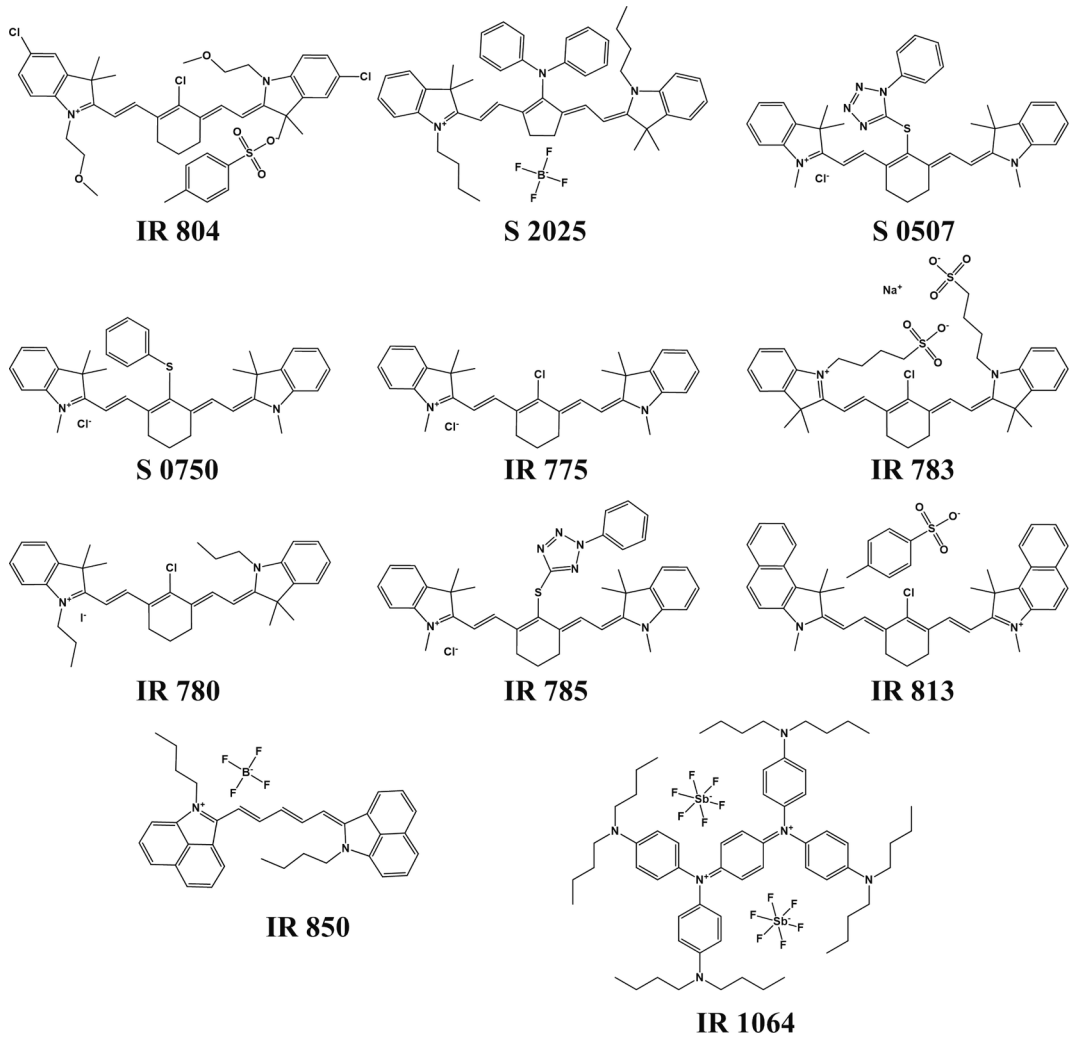


Fig. 3. Chemical structures of the NIR absorbing dyes used in this study (see also Table S1 for additional properties of these dyes).

2.5. Differential scanning calorimetry (DSC)

A Mettler-Toledo Differential Scanning Calorimeter (DSC) coupled to the Star software was employed to investigate the thermal behavior and properties of the cured polymers. The experimental protocol involved two heating cycles. The first cycle was utilized to calculate the conversion using the exotherms of polymerization using:

$$\textbf{\textit{DSC Conversion}}\ (\%) = \left[1 - \frac{\textbf{normalized integral of illuminated sample}}{\textbf{normalized integral of non} - \textbf{illuminated sample without dye}}\right]\boldsymbol{x}100 \quad (2)$$

while the second cycle aimed to determine the glass transition temperature (T_g) by drawing the tangent of the transition phase. During the analysis, a temperature ramp from 20 to 250 °C with a rise in temperature of 10 K.min^{-1}was employed. Based on reproducibility experiments the error bars on *Tg* was typically < 5°C.

2.6. Thermal imaging camera

Thermographic measurements were conducted using a FLUKE-TIX580/9HZ thermal camera. The sample was poured into mold with a depth of 1.4 mm and placed below the emitter of the laser diode. The video recording was switched on together with the light or 10 s before, see figures. The temperature was followed in real time for 300 s. Analysis took place using Fluke SmartView Classic 4.4 software. Based on reproducibility experiments the error bars was typically <5 °C (on the measured temperature during polymerization).

2.7. Uv–vis absorption spectra of NIR dyes

The absorption spectra of the NIR dyes were recorded in acetonitrile using a JASCO V750 spectrometer. Due to the variations in purity, the concentration was not absolutely quantitative, but was in the range of 5 10^{-6} mol/L.

2.8. Thermogravimetric experiments

The decomposition temperatures of NIR dyes were determined using the Mettler Toledo TGA DSC 3+ and the Star software. The samples were

Table 1
Characteristics of NIR heaters: maximal absorption wavelength (λ_{max}) in acetonitrile (5.10^{-6} mol/L) and degradation temperature measured by TGA. Full names and additional information can be found in Table S1.

NIR heaters	λ_{max} (nm)	ε [a] L·mol^{-1}·cm^{-1}	$T_{degradation}$ 10 % (°C)
S0804	788	96000	248
S2025	793	87000	267
S0507	793	12300	196
S0750	781	135000	283
IR775	768	119000	187
IR783	775	161000	n.m.
IR780	779	102000	198
IR785	810	23000	262
IR813	820	154000	283
IR850	852	28000[b]	232
IR1064	950	114000	288

n.m.: not measured.
[a] The purity of the compounds is partly not known, Table S1.
[b] Solibility problem: value given in toluene.

subjected to a temperature program ranging from 20 °C to 400 °C at a heating rate of 10 K.min^{-1}.

3. Results and discussion

3.1. Properties of the NIR dyes used as heaters

As key parameters for their use as heater, the light absorption properties of the different NIR dyes were investigated in acetonitrile with the concentration in the range of 5.10^{-6} mol.L^{-1} (Table1). Due to the differences in purity of the dyes which is for example only 90 % for IR783, we only present in this table a qualitative analysis of the dyes optical properties. Many of them exhibit a good absorption for the 760–850 nm range and only IR1064 exhibit a significant bathochromic shift. For efficient photothermal effect, a good matching between the absorption of the dye and emission spectrum of the light source is required. The selected light sources (see below) will allow efficient light excitation of the dyes.

To ensure a potential use as heater, their thermal stability must be also checked. The dye must be thermally stable to prevent its premature degradation and thus allowing sufficiently high temperatures to be reached. Therefore, the degradation temperatures of the NIR dyes, i.e. the temperature at which 10 % of the mass is lost, have been evaluated by TGA experiments. All the dyes (excepted IR783) exhibit a decomposition temperature higher than 180 °C; this latter temperature is high enough for epoxy/anhydride activation. Markedly, some dyes (IR813, S0750) exhibit very high decomposition temperature (close to 285 °C).

3.2. Proposed NIR photothermal mechanism

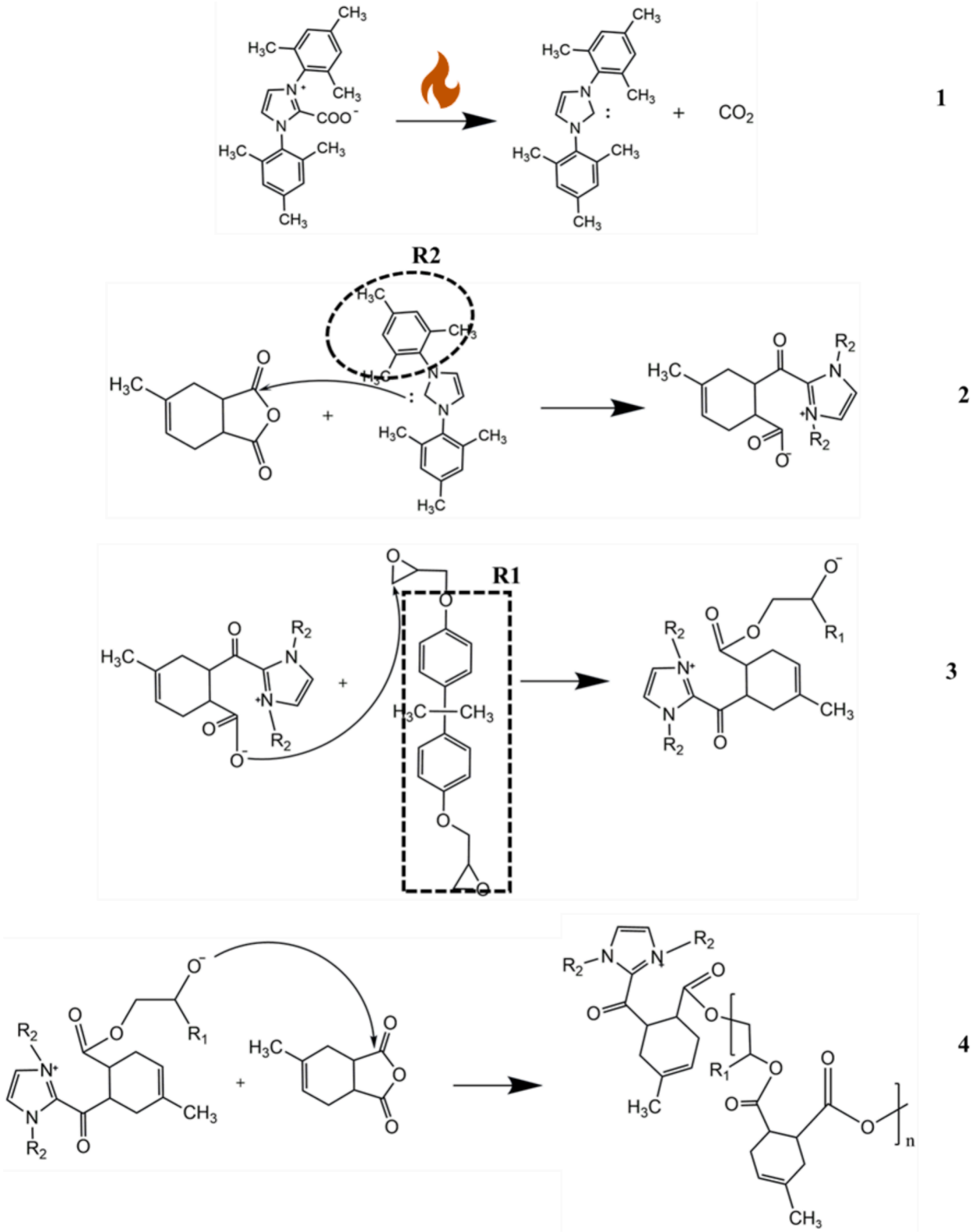
Markedly, upon few minutes of NIR irradiation of an epoxy/anhydride blend composed of BADGE (52 wt%) and THPA (48 wt%) in the presence of IR813 (0.1 phr) and NHC-G (0.5 phr), a hard material, namely a polyester thermoset, is obtained. No gelation/polymerization is observed without NIR dye or without NHC-G. These facts clearly prove that both compounds are mandatory for NIR induced polymerization. The proposed reaction mechanism, based on previous studies [22,23,24] resulting in the alternating copolymer; one unit from the anhydride followed by one unit from the epoxide, is presented in Scheme 1. Upon NIR light, the NIR dye through a photothermal effect (vibrational and collisional relaxation of the excited states) generates a huge increase of the temperature. Therefore, the N-Heterocyclic Carbene generator, under the influence of this heat, releases CO_2, which was detected by RT-FTIR by a characteristic peak at about 2360 cm^{-1} (See Fig. 4) [20,25], and generates NHC. But ideally the N-Heterocyclic Carbene released (1) subsequently attacks the anhydride (2), which can then induces the polymerization (3,4). This polyaddition consists in the alternating reaction with the epoxide (3) and the anhydride (4) resulting in the formation of polyester chains. Any kind of protonation (not shown), i.e. with water, will terminate the reaction. The elevated temperature not only favors the kinetic but also prevents termination by atmospheric humidity [26,27].

For the polymerization of epoxy-anhydride systems using NIR dye/NHC generator system upon NIR light, a change of the color formulation is found (see below). This result indicates the presence of side reactions during polymerization recognizable by a change of the dye structure. Indeed, this color change highlighted a partial NIR dye degradation that could be indicative of a shortening of the conjugation of its π system, indicated by a blue shift in light absorbance, a reduction in absorbance and a reduction in sample heating upon irradiation.

3.3. Monitoring of the NIR induced photothermal ROCOP of BADGE and THPA

In this section of the manuscript, we demonstrate the possibility of monitoring and evaluating the photothermal behavior as well as the possibility of polymerizing the resin system using the combination of a NIR heating dye and the NHC-G. The Fluke thermal camera was used for real-time thermal imaging to track the temperature on the sample surface (Fig. 5A). At the same time, the conversion of epoxy peaks in the near-infrared region (4470 to 4568 cm^{-1}) was monitored via FT-IR spectroscopy in transmission (i.e. the entire thickness was monitored, (Fig. 5 B)).

The photothermal performance of the systems (temperature reached vs. irradiation time) is shown in Fig. 5A using IR813 as heater and in Figure S1 using IR785. The conversion of epoxies during experiments performed under identical conditions are shown in Fig. 5B using IR813 as heater. The photothermal effect associated with the presence of the NIR heater can clearly be observed. Without a heating dye, with and without NHC-G, the temperature increase is very low (Fig. 5A, curves no. 3 and 4; T < 60 °C) without polymerization. The dye alone led to a high and rapid increase of the temperature of the sample without polymerization taking place. Polymerization was only observed in the presence of both IR813 and NHC-G (see Fig. 5B, curve 1 and part 3.2 above). Markedly, the temperature reached more than 150 °C in presence of the NIR dye and NHC-G. Simultaneously, the colors of the resin mixture were changing in general from green to yellow indicating a degradation of the NIR dye with a potential interruption of its conjugated π system. This effect was observed for all tested dyes, Figure S2. This change in color is not fully understood but surely caused by a shortening of the dye conjugated π-systems that could be associated with different side reactions e.g. redox reactions and/or reactions of the dye with the NHC released or with the different polymerization intermediates shown in Scheme 1. For IR785 as heater (Figure S1) in place of IR813 the temperature rose to 163 °C. This localized heat was found necessary for activating the thermal initiator that initiates the polymerization reaction without the need for other external heating. In presence of both IR813 and NHC-G (polymerization observed), the temperature reached was only around 10 degrees higher than with IR813 alone (no polymerization) (Fig. 5A curve 1 vs. curve 2). As depicted in Fig. 5B, curve 1, the presence of IR813 and NHC-G results in a rapid initiation of the conversion process upon irradiation (around 10 s). This conversion rate steadily increases, reaching approximately 70 % after 200 s. Conversely, the absence of IR813 (Fig. 5B, curve 2) significantly limits the conversion process, despite the generation of heat (as observed in Fig. 5A). This latter result suggests that the heat release by the photothermal effect (light to heat conversion effect induced by the NIR dye) is prominent in regards to the heat released by the exothermicity of polymerization.



Scheme 1. Proposed mechanism of the anionic alternating copolymerization of DGEBA and PTHA in the presence of NHC-G after the photothermal initiation. (1) heat-induced decarboxylation of the thermal initiator. (2) Initiation (3,4) alternating anionic polymerization.

3.4. Contrasting kinetics of NIR and thermal curing: Highlighting NHC-G's role

A striking contrast in polymerization kinetics was observed between the NIR-assisted and purely thermal curing processes (Fig. 6). The NIR-induced photopolymerization yielded ~70 % of conversion within a mere 300 s for the IR813 (in part 3.3 above). This substantial reduction in curing time underscores the efficacy of the NIR approach in accelerating the polymerization reaction. Furthermore, control experiments conducted without the NHC-G component revealed an absence of

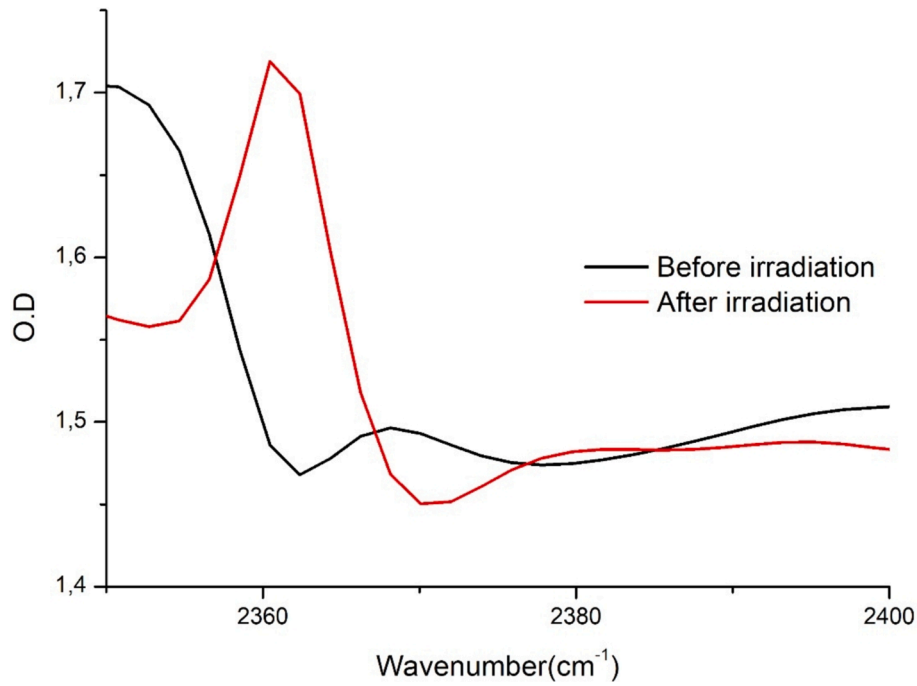


Fig. 4. Detection of CO_2 within RT-FTIR experiment for Epoxy/Anhydride blend (52 % /48 % (w/w)) in presence of 0.1 phr IR785 and 0.5 phr NHC-G upon NIR light irradiation by Laser Diode@785 nm (0.4 W/cm^2). Thickness = 1.4 mm under air.

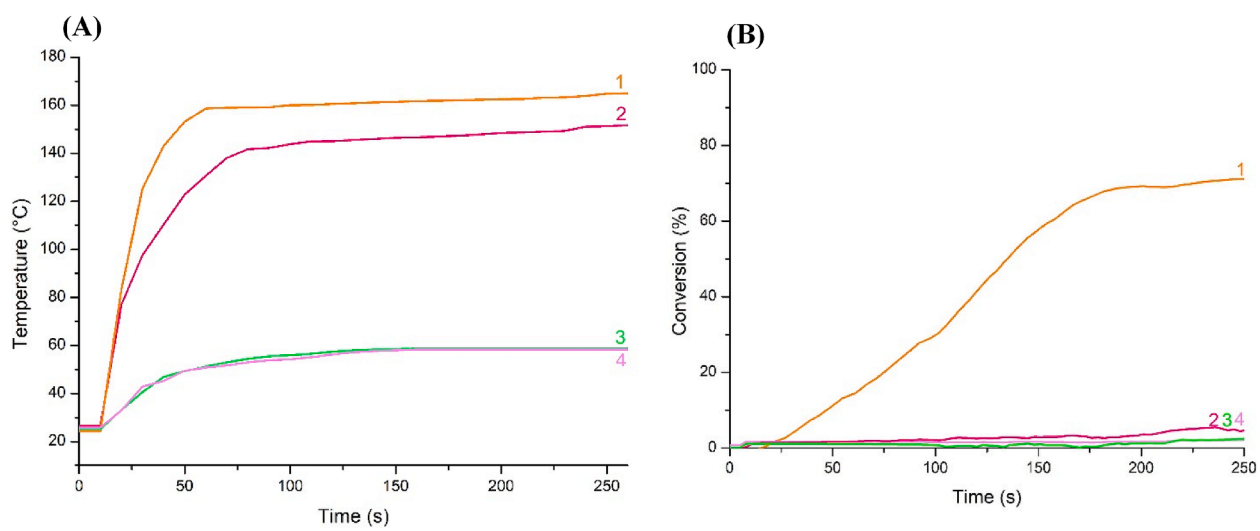


Fig. 5. (A) Thermal image analysis (temperature vs time) for Epoxy/Anhydride blend (52 % /48 % (w/w)) and corresponding/simultaneous (**B**) Photo-polymerization profiles (epoxy functions conversion in relation to irradiation time) for irradiation by Laser Diode@785 nm (0.4 W cm^{-2}) in presence of (1) 0.1 phr IR813 and 0.5 phr NHC-G; (2) 0.1 phr IR813; (3) 2 phr NHC-G; (4) without both IR813 and NHC-G. The irradiation starts at t = 10 s.

polymerization (Fig. 6A), even under prolonged thermal treatment. This observation strongly suggests that NHC-G plays a critical role in initiating the polymerization reaction (Fig. 6B), highlighting its indispensability in this system.

3.5. Stability of the different NIR dyes towards the NHC-G

In the initial phase of our investigation, we demonstrated the efficacy of NIR heaters in activating the photothermal effect through NIR light irradiation (e.g. using IR813 or IR785/NHC-G in part 3.3 above). However, a critical step for the photothermal polymerization is to confirm the stability of both the NIR dye and NHC-G in the epoxy/anhydride matrix. Specifically, NIR heaters must absorb NIR light and subsequently increase the temperature, thereby facilitating the activation process.

Moving forward, our subsequent investigation focused on assessing the stability of various dyes within the resin in the presence of NHC-G. Our findings, as depicted in Fig. 7, indicate noticeable changes in the coloration of the formulation without any recognizable gelling using IR850 and IR1064 after 12 h of stirring, without exposure to any non-ambient irradiation (the samples are fully kept in the dark). This change in color suggests a side-reaction and a degradation of the dye in presence of NHC-G. The change in color was never observed without the presence of NHC-G. The degradation of these two NIR dyes (IR850 and IR1064) was confirmed in photothermal experiments (Figure S2) with temperatures reached by these systems (consisting of Epoxy, Anhydride,

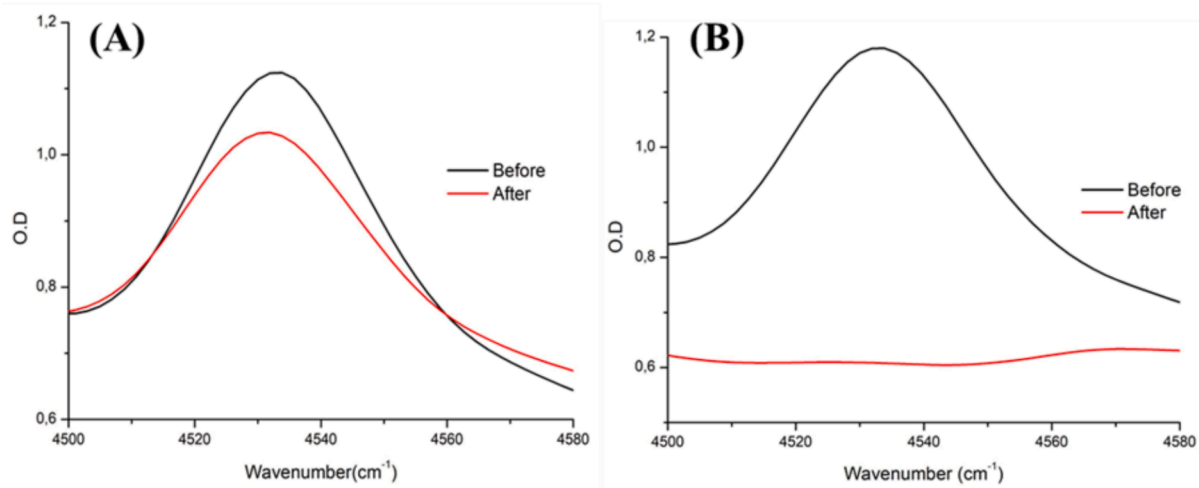


Fig. 6. Comparison of epoxy function within RT-FTIR experiment for Epoxy/Anhydride blend (52 % /48 % (w/w)) (A) without NHC-G and (B) in presence of 0.5 phr NHC-G before and after thermal polymerization for 20 min in 140 °C. The thickness of the samples was approximately 1.4 mm.

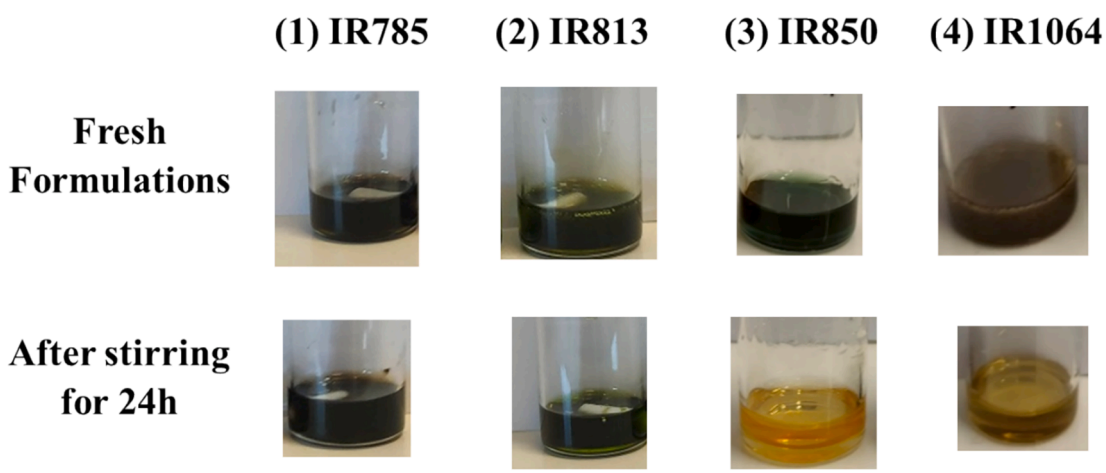


Fig. 7. Pictures of the formulations for Epoxy/Anhydride blend (52 % /48 % (w/w)) before and after 24 h of stirring in presence of (1) 0.1 phr IR785 and 0.5 phr NHC-G; (2) 0.1 phr IR813 and 0.5 phr NHC-G; (3) 0.1 phr IR850 and 0.5 phr NHC-G; (4) 0.1 phr IR1064 and 0.5 phr NHC-G.

dye, and NHC-G) remaining below 100 °C. By contrast, the same systems devoid of NHC-G reached a temperature of 140 °C. This disparity in temperature profiles has a significant impact on polymerization behaviors i.e. IR850 and IR1064 exhibit no polymerization, contrasting with the outcomes observed with the other dyes utilized in the initial phase of our study (IR813 and IR785). Except IR850 and IR1064, all the other investigated NIR dyes were found stable in presence of NHC-G, see next section, and can lead to sufficient photothermal effects upon NIR light to activate an efficient polymerization. This change in color for storage in the dark of IR850 and IR1064 based formulations is not fully understood but clearly highlight the dye degradation probably caused by a shortening of their conjugated π systems. This can be ascribed to different side reaction with the NHC-G at room temperature as these systems are color stable without NHC-G.

3.6. Structure/reactivity relationships for the different NIR dyes

In this section, only the stable systems were kinetically analyzed (Fig. 8 and Fig. 9). These are the resin mixtures with NIR dyes and NHC-G in which no significant color change was observed (for at least 48 h) during the mixing process, Figure S2.

The temperature evolutions of the epoxy/anhydride systems and the epoxy conversions (with and without NIR dyes and with and without NHC-G) are displayed in Figs. 8 and 9. As expected, in the absence of NIR heaters, the temperature of the resins remained low and the conversion was minimal under NIR irradiation, resulting in an absence of polymerization (Table 2).

Markedly, in the presence of the different NIR dyes and the NHC-G, the temperature increased, ranging from 130 °C for S0750 to 180 °C for S0804 (Table 2) and the dyes change their color in general from green to orange yellow (Figure S2). Efficient photothermal effects were observed in agreement with the good light absorption properties of these dyes at 785 nm. The Fig. 9 compares the photopolymerization performance of photosensitive resins formulated with different near-infrared dyes. Furthermore, formulations containing IR785 or IR813, demonstrated rapid polymerization kinetics. Markedly, the IR785-containing resin reached a conversion of 87 % in less than 160 s, while the IR813 formulation achieved 69 % conversion within the same timeframe. Following this initial rapid phase, both systems reached a plateau in conversion. However, other dyes resulted in lower epoxide conversions (~50–30 % after 300 s).

The presence of NIR dye and NHC-G was found mandatory to observe polymerization, however there is no clear correlation between the temperature reached and the final epoxy conversion, i.e. the three dyes with the best photothermal effect do not lead to the highest epoxy conversion. This suggests that additional factors like molar content of

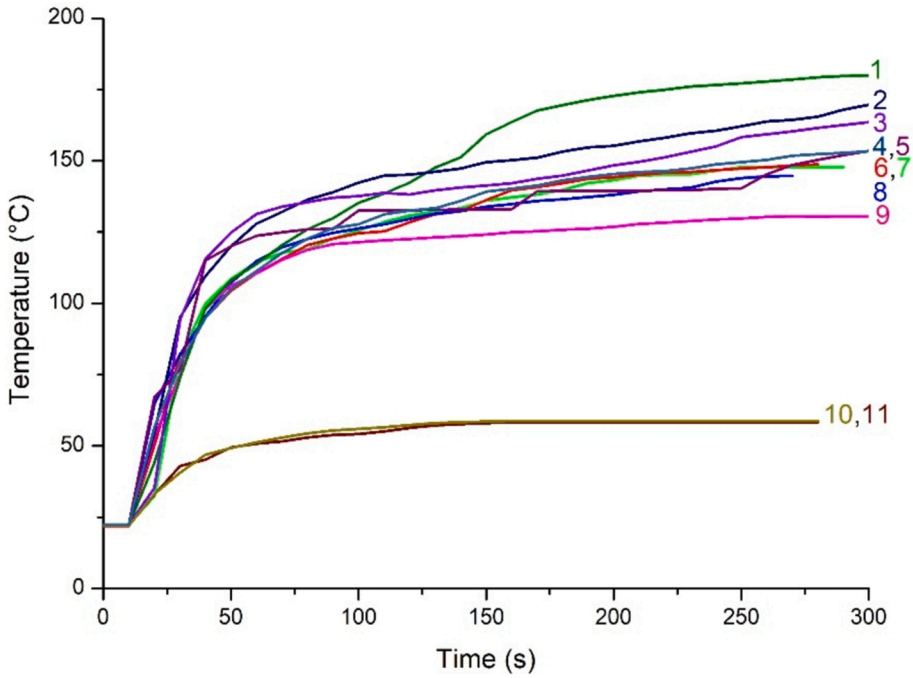


Fig. 8. Thermal imaging experiment (temperature vs time, 1.4 mm thick sample) for Epoxy/Anhydride blend (52 % /48 % (w/w)) irradiated with a Laser Diode@785 nm (0.4 W cm^{-2}) in presence of (1) 0.1 phr S0804 and 0.5 phr NHC-G; (2) 0.1 phr S2025 and 0.5 phr NHC-G; (3) 0.1 phr IR785 and 0.5 phr NHC-G; (4) 0.1 phr IR775 and 0.5 phr NHC-G; (5) 0.1 phr IR813 and 0.5 phr NHC-G; (6) 0.1 phr IR783 and 0.5 phr NHC-G; (7) 0.1 phr IR780 and 0.5 phr NHC-G; (8) 0.1 phr S0507 and 0.5 phr NHC-G; (9) 0.1 phr S0750 and 0.5 phr NHC-G; (10) without dye and with 2 phr NHC-G; (11) without dye and without NHC-G. The irradiation starts at t = 10 s.

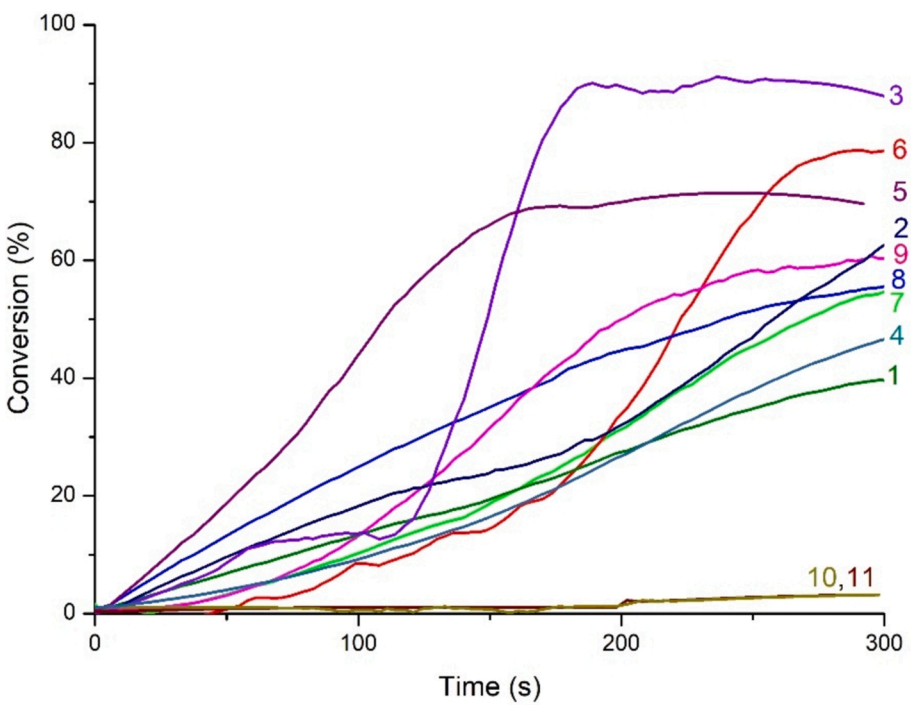


Fig. 9. Photopolymerization profiles (epoxide conversion vs irradiation time; 1.4 mm thick sample) for Epoxy/Anhydride blend (52 % /48 % (w/w)) for irradiation by Laser Diode@785 nm (0.4 W cm^{-2}) in presence of (1) 0.1 phr S0804 and 0.5 phr NHC-G; (2) 0.1 phr S2025 and 0.5 phr NHC-G; (3) 0.1 phr IR785 and 0.5 phr NHC-G; (4) 0.1 phr IR775 and 0.5 phr NHC-G; (5) 0.1 phr IR813 and 0.5 phr NHC-G; (6) 0.1 phr IR783 and 0.5 phr NHC-G; (7) 0.1 phr IR780 and 0.5 phr NHC-G; (8) 0.1 phr S0507 and 0.5 phr NHC-G; (9) 0.1 phr S0750 and 0.5 phr NHC-G; (10) 2 phr NHC-G; (11) without both dyes and NHC-G. The irradiation starts for t = 10 s.

the dye, purity, counter-ion, water content, solubility, the possible side reactions associated with the degradation of NIR dyes during polymerization as well as NIR dye/NHC-G interactions (and interactions upon light excitation e.g. photoinduced electron and energy transfer) could factor in the epoxy conversion. These complex behaviors are also reflected in the curves of the conversion profiles. To characterize these

Table 2
Characterization of NIR induced Anhydride/Epoxy polymerization using different NIR dyes and T_g of the corresponding obtained polymers.

Conditions 0.1 phr NIR dye and 0.5 phr NHC-G	maximal or plateau temperature reached (°C)	Epoxy Conversion in RT-FTIR (%)	Surface property	Conversion DSC (%) (After 48 h of storage of the irradiated samples)	T_g (°C) (Reheating cycle)
S0804 (1)	180	39	Tacky	67	101
S2025 (2)	171	62	Tack free	72	121
IR785 (3)	163	87	Tack free	93	122
IR775 (4)	155	46	Tacky	73	114
IR813 (5)	154	69	Tack free	75	107
IR783 (6)	148	79	Tack free	72	126
IR780 (7)	147	54	Tack free	77	120
S0507 (8)	144	55	Tack free	71	130
S0750 (9)	130	58	Tack free	87	131
without dye, with 2 phr NHC-G[a](10)	52	6	liquid	n.d.	n.d.
without dye and without NHC-G (11)	50	5	liquid	n.d.	n.d.

n.d. not determined.
[a] Non-illuminated samples were analyzed to calculate the conversion; Figure S4. The pure thermal treatment (20 min @140 °C) led to a Tg of 125 °C.

systems, we additionally conducted DSC measurements on the obtained polymer samples to determine the final extent of the reaction. The integration of the exothermic signal attributed to the polymerization exotherm, which is observed at about 140 °C, was thus measured for all polymer samples (Figures S3 and S4). Additionally, the integration of the exothermic signal for a non-illuminated resin constituted of the mixture of epoxy and anhydride and NHC-G was defined as the maximum enthalpy of reaction. In this way, we were able to calculate the conversions of the systems (eq. (2) presented above). The T_g values of the resulting polymers were also determined using a second heating cycle (after the thermal treatment of one cycle).

After NIR irradiation and 48 h of storage, the samples were subjected to DSC analysis. Comparing these results with the final epoxy conversions obtained through RT-FTIR, we observed an increase in conversion ranging from 6 % to 29 % as shown in Table 2. This significant enhancement led us to conclude that dark polymerization occurred. We assume that the end of the reactive chain, Scheme 1, for a sufficiently solidified sample, is not consumed and that a slow reaction occurs even if no new propagating anionic species are formed.

To confirm this dark curing, we performed the DSC measurement for one sample immediately after NIR irradiation and 48 h later, resulting in a significant change in conversion from 69 % to 75 %, as shown in Figure S5. The heating process initiated by photothermal polymerization, thus persisted even after switching off the NIR light.

The glass transition temperatures of the polymers prepared by photothermal polymerization in presence of NIR dye and NHC-G were found to be in the range of 100–130 °C (Table 2) e.g. with T_g for S0750 (No. 9 in Figs. 8 and 9) close to 130 °C that is quite similar to those of polymers prepared by pure thermal treatment (20 min @140 °C; Tg = 125 °C from this work). It should be noted that this sample is neither the one with the highest temperature reached during polymerization, nor the one with the highest initial conversion. It can be assumed that the NHC were slowly and continuously released and converted in the first step, which led to a high concentration of anionic propagation species that were very well stabilized by the dye, as can be seen from the strong post-curing of around 30 %. Differences in chain length and mechanical properties of the different samples heated with dyes could be possible, which we would like to investigate in future research using soluble polymers (simpler monomers).

4. Conclusion

The results conclusively demonstrate that photothermal activation of epoxy/anhydride ring-opening copolymerization using a thermolatent NHC generator, facilitated by NIR light, is possible, feasible and effective. While the NIR dyes absorb the incident light, it is the subsequent conversion of this light energy into heat (vibrational and collisional relaxation from their excited states) that provides the necessary activation energy/heat for the polymerization reaction to proceed.

The majority of the investigated dyes are stable for at least 48 h as a one compound (1 K) system in the epoxy/anhydride/NHC-G resins. Storage of the dye in one part of the resin and the NHC-G in a second cartridge is also possible for 2 K systems if the dye/NHC-G cannot be stored together. Due to complex effects, such as the heat-induced accelerating effect of dyes and their final products, the specific combinations of dyes and NHC-G induced by NIR light are challenging and require careful optimization of the formulation in future works. This innovative approach underlines the potential of NIR-induced polymerization for expanded application possibilities in material science, offering a promising alternative to conventional thermal and UV curing methods. Detailed mechanical investigations could not be carried out (lack of an area light source), so that it is currently unclear whether additional thermal treatments are necessary or not. In addition to the well-known application based on composites, and adhesives, this work opens the way to special applications for this chemistry such as the selective sealing of printed circuit boards (local heat effects, classical curing processes require temperature treatments “as high as >200 °C”), improved spatial control, deeper light penetration… Moreover, the ability to use safer wavelengths (such as UV) could prove beneficial in the future e.g. for 3D printing applications. The radiation induced reheating could also be beneficial for transesterification and “healing” of the material. The difference in activation, acceleration and dark curing, to name only a few is highly fascinating, interesting and challenging questions could be further investigated. For the presented cross-linked system in deep investigations are hardly possible. For such investigations simpler model system should be addressed and analyzed more in detail using for example mono-functional monomers (solvable polymers) and stochastic sampling for multi factor analysis (Machine Learing approaches). Other resins, NIR heaters or polymerization processes will be investigated by photothermal activation in forthcoming works. Interestingly, the incorporation of NIR heaters in polymerization systems lead to significant advancements in polymer technology, with the capacity to improve efficiency, precision, and safety in polymer production processes.

CRediT authorship contribution statement

Rania Rejeb: Writing – original draft, Data curation. **Philibert Lenormand:** Writing – review & editing, Data curation. **Gaspard Bichot:** Writing – review & editing, Data curation. **Didier Gigmes:**

Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Frédéric Dumur:** Writing – review & editing, Methodology, Investigation, Funding acquisition. **Michael Schmitt:** Writing – review & editing, Supervision, Investigation, Formal analysis, Conceptualization. **Julien Pinaud:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jacques Lalevée:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors thank the Agence Nationale de la Recherche for the funding of this work (ANR expo-photo; ANR-21-CE06-0037).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113589.

Data availability

Data will be made available on request.

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